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 Oefeningenreeks 6A nr. 7.5

$$\begin{aligned}
 &\text{Opgave: } x_1 \cdot x_3 = 36 \text{ en } x^2 + x^4 = 60 \\
 &\Rightarrow \begin{cases} x_1 \cdot (x_1 \cdot q^2) &= 36 \\ x_1 \cdot q + x_1 \cdot q^3 &= 36 \end{cases} \Rightarrow \begin{cases} (x_1 \cdot q)^2 &= 36 \\ x_1(q + q^3) &= 60 \end{cases} \\
 &\Rightarrow \begin{cases} x_1 \cdot q &= 6 \\ x_1(q + q^3) &= 60 \end{cases} \Rightarrow \begin{cases} x_1 &= \frac{6}{q} \\ x_1 \cdot (q + q^3) &= 60 \end{cases} \\
 &\Rightarrow \frac{6}{q}(q + q^3) = 60 \Rightarrow 6q^2 + 6 = 60 \Rightarrow q^2 + 1 = 10 \Rightarrow q^2 = 9 \\
 &\Rightarrow q = \pm 3 \Rightarrow x_1 = \frac{6}{q} = 2 \text{ of } x_1 = -2 \\
 &\Rightarrow x_2 = x_1 \cdot q = 6 \\
 &\Rightarrow x_3 = x_2 \cdot q = 18 \text{ of } x_3 = x_2 \cdot q = -18 \\
 &\Rightarrow (2, 6, 18, \dots) \text{ of } (-2, 6, -18, \dots)
 \end{aligned}$$

References